

Lab: Where Fiber Meets World

Materials Exploration — Systems

Objective

Explore yarn as a system by directly investigating how fiber composition, twist, ply, and environmental context shape yarn behavior. You will build firsthand understanding of internal states, boundaries, and environmental influences.

Materials Needed

- 3-5 yarn samples of different fiber types (e.g., 100% wool, 100% cotton, 100% acrylic, a silk blend, a synthetic blend)
- A crochet hook in an appropriate size for the yarns
- A small bowl of warm water
- A small bowl of cool water
- A ruler or tape measure
- A notebook for observations
- Optional: a magnifying glass or phone camera with zoom

Part 1: Mapping Internal States (20 minutes)

For each yarn sample, investigate and record the following internal states:

1. **Fiber content:** Read the label. What fibers are present?
2. **Twist direction:** Hold the yarn vertically and observe — does the twist spiral up to the left (S-twist) or to the right (Z-twist)?
3. **Twist tightness:** Gently pull a short length between your hands. How much does it resist untwisting? Rate it 1-5.
4. **Ply count:** Carefully untwist a small section. How many individual strands are plied together?

5. **Surface texture:** Run the yarn through your fingers. Is it smooth, fuzzy, hairy, bumpy, slippery? These surface qualities are the boundary of the system.

Record your findings in a table:

Sample	Fiber	Twist Dir.	Twist Tightness (1-5)	Ply	Surface

Part 2: Testing the Boundary (20 minutes)

Now explore how each yarn system interacts with its environment at the boundary:

1. **Moisture test:** Dip a 6-inch length of each yarn in warm water for 30 seconds. Remove it and observe:
2. How quickly did it absorb water? (Wool and cotton absorb; acrylic resists.)
3. How does the texture change when wet?
4. How heavy does it feel compared to dry?
5. **Stretch test:** Gently pull a dry length of each yarn. How much does it stretch before resisting? Now do the same with the wet sample. Does moisture change the stretch?
6. **Recovery test:** Stretch each yarn to its comfortable limit, then release. Does it spring back (like wool) or stay stretched (like cotton)?

Part 3: Environmental Context (15 minutes)

1. Crochet a small 10-stitch, 5-row swatch with one yarn sample.
2. Measure the width and height of the swatch.
3. Now dampen the swatch in warm water and lay it flat to dry. This is a mini-blocking.
4. After it dries, measure again. How did the environment (water + drying) change the system?

Part 4: The Dye Lot Question (10 minutes)

If you have two skeins of the same yarn from different dye lots (or can find examples online):

1. Hold them side by side in natural light. Can you see any difference?
2. Now alternate rows crocheting with each. Does the difference become more or less visible in the fabric?
3. Reflect: What does this tell you about how subtle differences in manufacturing environment become visible when the system is put to use?

Discussion Questions

1. Which of your yarn samples had the most "active" boundary — the most noticeable interaction between internal states and environment?
2. Did any yarn surprise you? Was there a gap between what the label predicted and what you experienced?
3. How would you now describe a skein of yarn to someone using the language of systems, boundaries, and states?

Reflection

Write a short paragraph connecting your hands-on findings to the concept of a Markov blanket. Where exactly is the boundary of the yarn system? What information passes through it? What stays hidden inside?